

Lab 08

Due March 28, 5:00 PM PST

AUTHOR

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Modify this template file to create your submission.

Problem 1a

```
## write your answer here
library(tidytext)
library(janeaustenr)
library(stringr)
library(dplyr)
library(ggplot2)

book_words <- austen_books() |>
  unnest_tokens(word, text)

sense_words <- book_words |>
  filter(book == "Sense & Sensibility") |>
  mutate(long_word = str_length(word) >= 10) |>
  summarize(proportion = mean(long_word))

sense_words
```

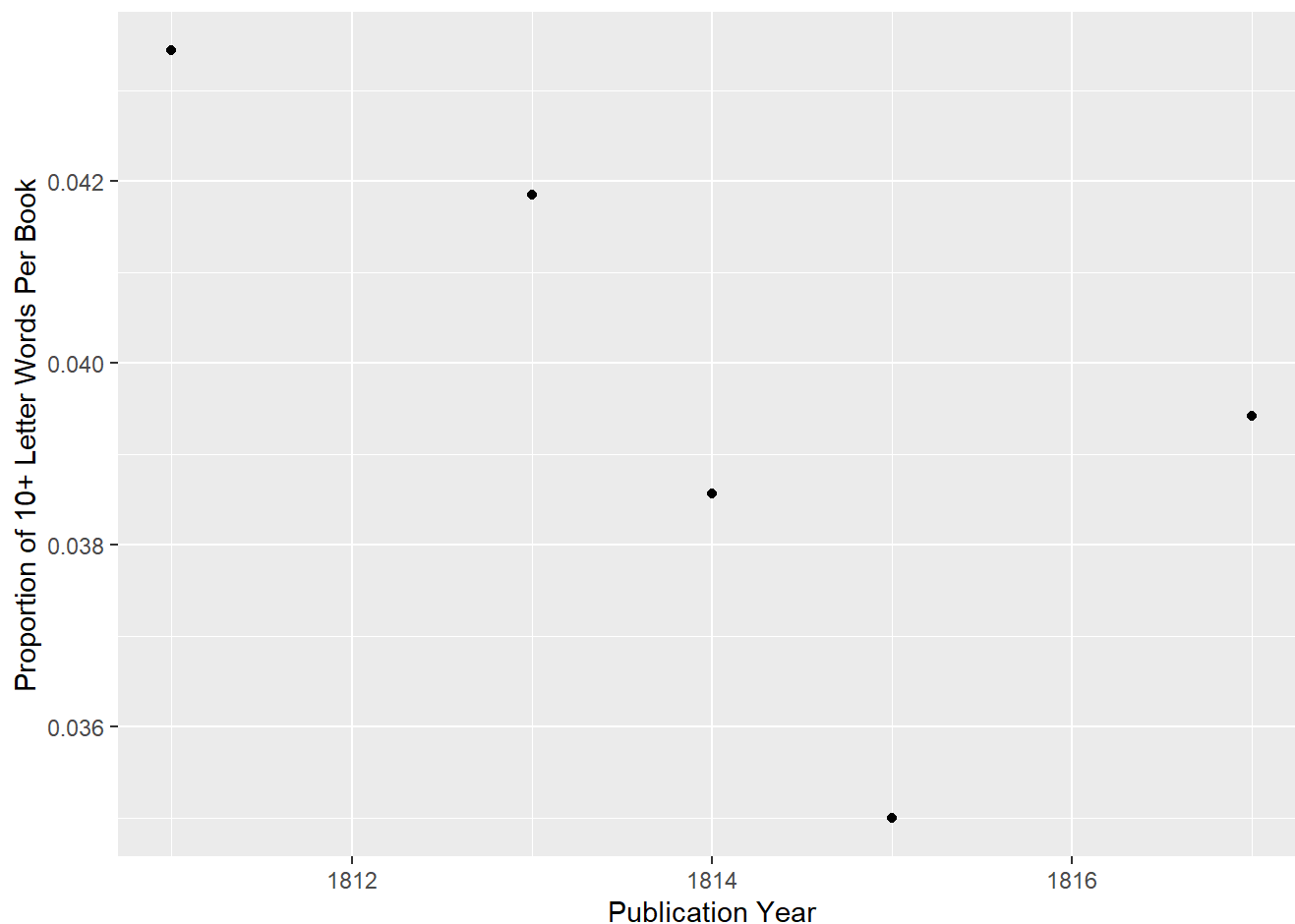
```
# A tibble: 1 × 1
  proportion
    <dbl>
1      0.0434
```

Problem 1b

```
## write answer here
book_years <- tibble(
  book = c("Sense & Sensibility", "Pride & Prejudice", "Mansfield Park",
    "Emma", "Northanger Abbey", "Persuasion"),
  year = c(1811, 1813, 1814, 1815, 1817, 1817)
)

book_proportions <- book_words |>
  group_by(book) |>
  summarize(proportion = mean(str_length(word) >= 10)) |>
  left_join(book_years, by = "book")
```

```
ggplot(book_proportions, aes(x = year, y = proportion)) +  
  geom_point() +  
  labs(x = "Publication Year", y = "Proportion of 10+ Letter Words Per Book")
```



Problem 1c

```
## write answer here  
seven_words <- book_words |>  
  mutate(word_length = str_length(word)) |>  
  filter(word_length == 7) |>  
  filter(str_detect(word, "^[wxyz]"), str_detect(word, "e$")) |>  
  distinct(word) |>  
  arrange(word)
```

```
seven_words
```

```
# A tibble: 4 × 1
```

```
  word  
  <chr>  
1 warfare
```

```
2 welcome  
3 welfare  
4 wimpole
```

```
seven_words_count <- nrow(seven_words)  
seven_words_count
```

```
[1] 4
```